how the molecules were distributed. Focusing on big-picture statistical features— is the gas spread out or bunched up, that is, does it have high or low entropy?—is far more illuminating.

Entropy, the Second Law, and the Arrow of Time

The tendency of physical systems to evolve toward states of higher entropy is known as the *second law of thermodynamics.* (The first law is the familiar conservation of energy.) As above, the basis of the law is simple statistical reasoning: there are more ways for a system to have higher entropy, and "more ways" means it is more likely that a system will evolve into one of these high-entropy configurations. Notice, though, that this is not a law in the conventional sense since, although such events are rare and unlikely, something *can* go from a state of high entropy to one of lower entropy. When you toss a jumbled stack of pages into the air and then gather them into a neat pile, they *can* turn out to be in perfect numerical order. You wouldn't want to place a high wager on its happening, but it *is* possible. It is also possible that the bumping and jostling will be just right to cause all the dispersed carbon dioxide molecules to move in concert and swoosh back into your open bottle of Coke. Don't hold your breath waiting for this outcome either, but it can happen.[10]

The large number of pages in *War and Peace* and the large number of gas molecules in the room are what makes the entropy difference between the disordered and ordered arrangements so huge, and what causes low-entropy outcomes to be so terribly unlikely. If you tossed only two double-sided pages in the air over and over again, you'd find that they landed in the correct order about 12.5 percent of the time. With three pages this would drop to about 2 percent of the tosses, with four pages it's about .3 percent, with five pages it's about .03 percent, with six pages it's about .002 percent, with ten pages it's .000000027 percent, and with 693 pages the percentage of tosses that would yield the correct order is so small—it involves so many zeros after the decimal point—that I've been convinced by the publisher not to use another page to write it out explicitly. Similarly, if you dropped only two gas molecules side by side into an empty Coke bottle, you'd find that at room temperature their random motion would bring them back together